Photometric Redshift Prediction Using Multiple Linear Regression on SDSS Data

---

Dataset: SDSS · 99,992 observations after cleaning
Model: Multiple Linear Regression (MLR) · 5-band
Target: Cosmological Redshift z
Classes: GALAXY · QSO · STAR
Tools: R · Python/rpy2 · Google Colab · GitHub Pages
Author: Konton_Chan
Seed / Split: 64 | 80/20 Train/Test

Github : https://github.com/Konton-Chan/MLR_Redshift_Using_R

1. Introduction

Cosmological redshift (z) measures how much the universe has expanded since light left a distant source. Spectroscopic measurement is precise but telescope-time-expensive. Photometric redshift estimation — deriving z from broadband magnitudes — provides a scalable complement for large sky surveys.

This project applies MLR to the Sloan Digital Sky Survey (SDSS) catalog to predict z for GALAXY, QSO, and STAR objects. The full pipeline runs in R inside Google Colab; trained models are served as a browser-based interactive tool on GitHub Pages.

1.1 Objectives

- Exploratory Data Analysis (EDA) on 100,000 SDSS photometric objects.
- Physics-informed outlier removal — keep false positives, delete true errors.
- Train & evaluate 8 MLR models (4 datasets × 2 feature sets).
- Select best model per class using R^2, RMSE, MAE, SMAPE.
- Export coefficients + Prediction Interval parameters to JSON (8-digit precision).
- Deploy as an interactive web application with spectral simulations.

1.2 Scope & Tools

Predictors are limited to the five SDSS photometric bands (u, g, r, i, z). SDSS spectroscopic redshift is the ground-truth target. All models use random seed 64 and an 80/20 train-test split.

Tools: R (lm, jsonlite), Python (rpy2, pandas), Google Colab, HTML/CSS/Vanilla JS, GitHub Pages.

2. Background Theory

2.1 Cosmological Redshift

When the universe expands, photon wavelengths are stretched. The fractional shift is defined as:

$$z = (\lambda_obs - \lambda_rest) / \lambda_rest$$

For galaxies $z > 0$ (recession); $z = 0$ (at rest); $z < 0$ (blueshift, approaching). Stars in the Milky Way show only tiny Doppler shifts ($z \approx 0$), while distant quasars can exceed $z = 7$.

2.2 SDSS Photometric Bands

Band	Central Wavelength	Spectral Region	Typical Range
u	3551 Å	Ultraviolet	15 – 24 mag
g	4686 Å	Blue / Green	14 – 24 mag
r	6165 Å	Red	13 – 24 mag
i	7481 Å	Near Infrared	12 – 23 mag
z	8931 Å	Infrared	12 – 23 mag

2.3 Multiple Linear Regression

MLR expresses the target as a linear combination of predictors:

$$z = b0 + b1 \cdot u + b2 \cdot g + b3 \cdot r + b4 \cdot i + b5 \cdot z_band + \varepsilon$$

Model quality is assessed with R^2 (explained variance), RMSE, MAE, and SMAPE. SMAPE replaces traditional MAPE to handle near-zero redshift values common in STAR observations that would otherwise cause MAPE $\rightarrow \infty$.

2.4 Prediction Interval (PI) vs Confidence Interval (CI)

A CI quantifies uncertainty in the mean prediction. A PI additionally covers individual-observation variability — the statistically correct choice when predicting the redshift of a single new object:

$$PI = pred \pm t(0.975, df) \times \sigma \times \sqrt{(1 + 1/n)}$$

where σ = residual standard error, df = n_train − p − 1.

This is exactly R's

```
predict(..., interval = "prediction", level = 0.95)
```

3. Dataset

3.1 Source

Data from the Sloan Digital Sky Survey (SDSS) — a wide-field optical imaging and spectroscopic survey covering roughly one-third of the sky. The raw file contains 100,000 rows × 18 columns.

Column	Type	Description
obj_ID	numeric	SDSS object identifier (not used in 5-band model)
alpha / delta	numeric	Right Ascension / Declination (degrees)
u, g, r, i, z	numeric	Photometric band magnitudes [10–30] → predictors
class	categorical	GALAXY / QSO / STAR
redshift	numeric	Target variable — spectroscopic redshift z
run_ID … fiber_ID	numeric	SDSS survey metadata (used only in MLR_all models)

3.2 Class Distribution (After Cleaning)

Class	Rows	Share	Redshift Range
GALAXY	59,443	59.4%	0.000 – 0.817
STAR	21,591	21.6%	≈ 0 (Doppler only)
QSO	18,958	19.0%	0.065 – 7.011
Total	99,992	100.0%	—

4. Data Preprocessing

4.1 EDA on Raw Data

Boxplots were generated for all 17 numeric features. An Interactive Row Filter confirmed 1 row with z = −9999 (sensor error flag). IQR-based outlier detection flagged 391 rows across the five bands.

4.2 Outlier Classification

Physical limits: valid magnitudes must satisfy $10 \leq \text{mag} \leq 30$. IQR outliers were classified as True Errors (physically invalid) or False Positives (statistically extreme but real bright/faint objects).

Band	IQR Fence Lower	IQR Fence Upper	IQR Outliers	False Pos ↓	False Pos ↑	True Errors
u	15.35	28.69	56	44	9	3
g	14.23	26.86	99	76	20	3
r	13.59	25.95	132	117	14	1
i	13.06	25.18	198	163	30	5
z	12.65	24.58	320	255	63	2

4.3 Cleaning Decision

Only the 8 True Error rows were removed. False positives are physically real objects (e.g., nearby bright galaxies) and were retained. Final cleaned dataset: 99,992 rows.

***Note:** Removed: Row 1 — QSO, i = 30.15 (sensor ceiling exceeded); Row 2 — STAR, r/i/z < 10 (detector saturation at 3 bands).

4.4 Post-Cleaning EDA

Applied to all four datasets (All, GALAXY, QSO, STAR): boxplots, full descriptive statistics (mean, SD, min, max, IQR, skewness, kurtosis, Pearson correlation, CV, Z-score, percentiles), histograms, correlation heatmaps, and scatter plots of magnitude vs. redshift.

5. Modeling

5.1 Train / Test Split

Dataset	Train Rows	Test Rows	Total
dataset_clean	79,993	19,999	99,992
dataset_GALAXY	47,554	11,889	59,443
dataset_QSO	15,166	3,792	18,958
dataset_STAR	17,272	4,319	21,591

5.2 Model Variants

Two formulas per dataset → 8 models total:

- MLR_*_all — 13 predictors: alpha, delta, u, g, r, i, z, run_ID, cam_col, field_ID, plate, MJD, fiber_ID
- MLR_*_5band — 5 predictors: u, g, r, i, z (photometric bands only)

5.3 Evaluation Metrics

Each model evaluated on Train and Test sets with R^2, Adjusted R^2, RMSE, MAE, and SMAPE. SMAPE is used because MAPE diverges when actual redshift ≈ 0 (STAR class, low-z galaxies):

$$\text{SMAPE} = \text{mean}\left(2 \cdot |\text{actual} - \text{predicted}| \,/\, (|\text{actual}| + |\text{predicted}|) \times 100 \right)$$

***Note:** Before fix: MLR_clean MAPE = 318,566% (undefined). After SMAPE: 83% (interpretable).

5.4 Results — Test Set (All 8 Models)

Model	Dataset	R^2	Adj R^2	RMSE	MAE	SMAPE
MLR_clean_all	Dataset_ clean	0.2898	0.2894	0.6102	0.3624	83.3%
MLR_clean _5band	Dataset_ clean	0.2856	0.2855	0.6135	0.3654	82.4%
MLR_GALAXY_all	Dataset_ GALAXY	0.7488	0.7485	0.1319	0.0812	29.5%
MLR_GALAXY_5band ✓	Dataset_ GALAXY	0.7246	0.7245	0.1354	0.0834	30.2%
MLR_QSO_all	Dataset_ QSO	0.1438	0.1409	0.8521	0.6084	38.2%
MLR_QSO _5band ✓	Dataset_ QSO	0.1436	0.1430	0.8522	0.6098	38.5%
MLR_STAR_all	Dataset_ STAR	0.0612	0.0583	0.000446	0.000277	127%
MLR_STAR _5band ✓	Dataset_ STAR	0.0125	0.0113	0.000457	0.000284	—

5.5 Model Selection Rationale

The _5band model is preferred over _all because the R^2 gain from adding 8 extra metadata features is below the threshold of 0.02 for GALAXY (gain = 0.024), and near zero for QSO and STAR. The 5-band model is simpler, more interpretable, and directly applicable to raw photometric catalogs without survey metadata.

***Note — STAR:** $R^2 \approx 0$ is physically expected. Stars in the Milky Way have no cosmological redshift; any z measurement reflects only tiny Doppler shifts ($z \approx 0.000$–0.001). RMSE = 0.000446 confirms the model predicts $z \approx 0$ accurately despite low R^2.

6. 95% Prediction Interval

6.1 Formula

The 95% PI for a single new observation is computed in R (Colab Cell 26) and exported to JSON:

$$PI = pred \pm t(0.975, df) \times \sigma \times \sqrt{(1 + 1/n)}$$

where σ = residual SE, df = n_train − p − 1, n = training size. The half-width is stored as `pi_half_width` in the JSON file so the browser can display it without recomputing.

6.2 PI Parameters — MLR_GALAXY_5band

Parameter	Value
n_train	47,554
Residual SE (σ)	0.13920465
df_residual	47,548
t-crit t(0.975, df)	1.96001388
PI half-width	0.27284592

6.3 CI vs PI — Example Prediction

Input: u = 20.14, g = 18.32, r = 17.41, i = 16.93, z = 16.60 (GALAXY)

Predicted z = 0.159082

Interval	Lower	Upper	Width	Note
CI 95%	0.157079	0.161085	0.004006	Mean estimate uncertainty
PI 95%	−0.113769	0.431932	0.545701	Single-object range

PI is ~136× wider than CI.

6.4 Negative Lower Bound

PI lower = −0.114 is a valid statistical result. Physically, $z < 0$ means blueshift (object approaching), which is not a model error but reflects uncertainty at low redshift. The SDSS training range is $z \geq 0$, so the bound is interpreted as $z \approx 0$ in practice. The web application shows this as a physics note.

7. Web Application

7.1 Architecture

Static single-page app on GitHub Pages:
https://konton-chan.github.io/MLR_Redshift_Using_R/
Prediction runs entirely in-browser using pre-exported JSON files.

File	Lines	Purpose
index.html	311	UI layout — all HTML structure & element IDs
script.js	1,139	Model loading, prediction, 4 simulation canvases, Doppler
style.css	514	Dark space theme, CSS variables, responsive breakpoints
MLR_*.json	—	Coefficients + PI params (3 files: GALAXY, QSO, STAR)

7.2 JSON Model Format

Each model exported from R Cell 26 at 8-decimal precision:

json

```
{ "model_name": "MLR_GALAXY_5band",
  "metrics": {
      "r2": 0.72461400, "rmse": 0.13541100,
      "residual_se": 0.13920465, "df_residual": 47548,
      "t_crit_95": 1.96001388, "pi_half_width": 0.27284592
  },
  "coefficients": {
     "b0_intercept": -1.78174828, "b1_u": -0.00620087,
     "b2_g": 0.02393263, "b3_r": 0.14824592,
     "b4_i": -0.01452566, "b5_z": -0.04263688
  }
}
```

7.3 Key Features

- Step 01 — Class selection: Loads the correct JSON model for GALAXY / QSO / STAR.
- Step 02 — Band input: u, g, r, i, z (10–30 per band; validated on input).
- Predicted z + equation: Animated counter + full MLR equation with substituted values.
- 95% PI box: Visual bar: Lower | track | Upper, Width chip, physics note if lower < 0.
- ① Doppler Colour Shift: Interactive slider over the PI range; spectral strip shows Hα colour.
- ② SED Profile: Magnitude bar chart across 5 SDSS bands for your object.
- ③ Spectral Line Shift: Emission lines (Ly-α, Hβ, Hα, …) shifted to observed wavelengths.
- ④ Cosmic Distance: Recession velocity, comoving distance (Mpc/Gly), light travel time.
- Model Performance cards: R^2 bars for all three classes in the info section.

8. Discussion

8.1 Why GALAXY Performs Best ($R^2 = 0.72$)

GALAXY redshifts occupy z = 0–0.8, where the SDSS bands sample well-known spectral landmarks (4000 Å break, Balmer series). In this range the colour-redshift relation is roughly linear, which suits MLR well. Only 5 predictors are enough to explain 72% of the variance.

8.2 Why QSO Performs Poorly ($R^2 = 0.14$)

Quasars span z = 0–7 with variable, broad emission lines. The same photometric band samples entirely different spectral features at different redshifts, violating MLR's linearity assumption. Non-parametric or deep-learning approaches are more appropriate for QSO photo-z estimation.

8.3 SMAPE vs MAPE

Standard MAPE divides by the actual value, exploding when actual ≈ 0 (STAR, low-z galaxies). SMAPE divides by the sum of actual and predicted, bounding the metric to [0%, 200%]. Results after the switch: GALAXY ≈ 30%, QSO ≈ 38%, STAR ≈ 127% (expected for near-zero targets; interpret alongside RMSE = 0.000446).

8.4 Limitations

- MLR cannot capture non-linear redshift-colour relations at high z.
- 5-band model ignores position and survey metadata that could improve accuracy.
- Training depth limited to SDSS imaging; very faint/bright objects may extrapolate poorly.
- STAR is structurally ill-suited for photometric redshift estimation.

9. Conclusion

Multiple Linear Regression with only five SDSS photometric magnitudes achieves $R^2 = 0.72$ for GALAXY redshift prediction on 11,889 test objects. The end-to-end pipeline covers raw ingestion, physics-informed cleaning, full EDA, training/evaluation across 8 models, and browser-based deployment with interactive spectral simulations.

Prediction Intervals — rather than Confidence Intervals — are reported to users, ensuring the uncertainty is statistically appropriate for single-object predictions. The Doppler colour strip provides intuitive physical context for non-specialists.

Future Work:

- Random Forest / Gradient Boosting for non-linear photo-z estimation.
- CNN on raw SDSS image cutouts for feature-free deep photo-z.
- Extend to fainter objects via deeper surveys (DES, Rubin/LSST).
- Bayesian or conformal prediction for full uncertainty quantification.

References:

[1] York, D.G. et al. (2000). The Sloan Digital Sky Survey. AJ, 120(3), 1579.

[2] Montgomery, D.C., Peck, E.A., Vining, G.G. (2012). Introduction to Linear Regression Analysis, 5th ed. Wiley.

[3] Collister, A.A. & Lahav, O. (2004). ANNz: photo-z with ANNs. PASP, 116, 345.

[4] Budavari, T. & Szalay, A.S. (2008). Probabilistic photometric redshifts. ApJ, 679, 301.

[5] SDSS Science Archive Server: https://www.sdss.org/

[6] R Core Team (2023). R: A language and environment for statistical computing.

[7] Konton_Chan. MLR_Redshift_Using_R. https://github.com/Konton-Chan/MLR_Redshift_Using_R